

Excellent ($\geq 90\%$)	Good (75-89%)	Satisfactory (50-74%)	Needs Improvement (30-49%)	Poor ($< 30\%$)
Code is fully functional and meets all the requirements of the practical. It runs without errors and produces the expected results.	Code is mostly functional but may have minor errors or missing functionality.	Code is functional in theory if all code bugs can be fixed. The students should demonstrate that they know the directions to fix the errors in their code.	Code has multiple errors or missing functions that significantly impact its functionality.	Code does not function properly and does not meet the requirements of the practical.

- The difficulty level of each milestone can be rated by

: Easy : Standard
: Hard : Challenging

- Some milestones have other milestones as pre-requisites – you cannot be awarded marks for their completion until you have completed these other milestones.

#	Pre-reqs	Milestone	
1		Initialise new sample set with sample count = 0, and set total weight of the new particle set to zero	1 mark 
2		calculate $w_diff = 1 - w_fast / w_slow$, and if $w_diff < 0$, set it to zero	1 mark 
3		Generate uniformly distributed samples according to a probability of w_diff	2 marks  
Complete function <code>sensorFunction(*)</code> - Total 6 Marks			
4.1		Calculate the cumulative importance weight here for resampling purpose, and save to the array <code>cd_old_set</code>	2 marks  
4.2	4.1	Generate samples based on weight	4 marks    
5	3 4.2	Allocate weights to new particles, and calculate total weights	2 marks  
6	3 4.2	Compute the value of “M”, i.e., the number of samples that the KL distance between particles and real distribution is less than “epsilon” with probability of “1 - delta”	4 marks    
7	3 4.2 6	Check whether number of samples in the new particle set is larger than M, if yes, break the resampling.	1 mark 
8	3 4.2 6 7	Normalised the weights in the new particle set	1 mark 

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